

Algebraic and Transcendental functions

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Algebra skills

Solve the following equations,

1. $-7 + \frac{1}{3} \log_2(x-6) = -6$
2. $3^{x+3} + 3^x = 81$

Use the properties to give the exact numeric value of the following logarithmic expression without using calculator, $\log_2(16)$.

Function analysis

Find the asymptote, critical point, domain and range of the following exponential and logarithmic functions,

1. $f(x) = -\log_2(x-3) + 5$
2. $g(x) = e^{-x+1} + 1$

Modelling with math

Population growth The population of a certain city is given by $P(t) = 234e^{0.071t}$, where $P(t)$ is measured in thousands of people and t is in years from the year 2000.

1. Find the population for the year 2014.
2. Find the time at which the population has tripled.
3. Find the *Half-life* value for this population.

Algebra Toolkit Algebra properties

$$a + b = b + a,$$

$$(a + b) + c = a + (b + c),$$

$$a(b + c) = ab + ac,$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

$$\frac{-a}{b} = -\frac{a}{b} = \frac{a}{-b},$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a} = a^{1/n}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$a^2 - b^2 = (a - b)(a + b)$$

Laws of exponents

$$a^r \cdot a^s = a^{r+s}, \quad (a^r)^s = a^{rs}$$

$$\frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r$$

$$\left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \quad b \neq 0$$

Laws of Logarithms

$$\log_a(AB) = \log_a(A) + \log_a(B)$$

$$\log_a\left(\frac{A}{B}\right) = \log_a(A) - \log_a(B)$$

$$\log_a(A^C) = C \log_a(A)$$

Base change formula

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$$

Inverse functions property

$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x)$$

$$\log_a(a^x) = x = a^{\log_a(x)}$$

Conic sections

Identify all the parameters¹ of the following conic sections. Express the conic section in the standard form.

1. $2x^2 + 2y^2 - 12x - 20y + 34 = 0$

2. $x^2 - 2y^2 - 2x + 12y + 19 = 0$

Conic Sections

Standard form of a circle

$$(x-h)^2 + (y-k)^2 = r^2$$

Standard form of a parabola

$$(x-h) = 4a(y-k)^2, \text{ or } (y-k) = 4a(x-h)^2$$

Standard form of an ellipse

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Standard form of a hyperbola

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

¹ Center, foci, vertex, asymptotes, radius, semi-axis. Depending on the conic section.